

Synthure: Four Interconnected Portals from One Clinical Note

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Abstract

Synthure is a multi agent clinical AI demonstrator that turns one clinical note into four interconnected stakeholder portals, one each for the patient, the physician, the hospital revenue cycle team, and the employer benefits team. The four portals share a single mutable encounter, so an action in one portal (approving a prior authorization, toggling a code, applying for financial assistance, submitting a claim) ripples through the others in real time and recomputes the shared risk, cost, and reimbursement figures.

A note is processed by a pipeline of agents: format validation, biomedical named entity recognition (NER), knowledge resolution of codes to labels, deterministic risk scoring, four grounded writer agents, and a verification and orchestration pass. When an Anthropic API key is present, the NER and writer agents are live Claude calls with forced tool_use schemas; without a key, a deterministic engine runs the same pipeline so the demo always works.

This report is deliberately transparent about what has and has not been measured. We report a real evaluation of the NER stage on a public labelled dataset: Claude based NER recovers the correct ICD 10 category in 60.4% of cases and the exact code in 23.5% ($N = 149$), against 0.0% for the deterministic regex extractor on the same prose inputs. The denial, retrieval, and readmission components are not yet benchmarked; rather than report numbers we cannot reproduce, we describe a concrete evaluation plan for each.

1 Introduction

The United States healthcare system allocates a large fraction of clinical capacity to administrative rather than patient facing work. Physicians spend a substantial share of their time on prior authorization, claims submission, denial management, and documentation, a pattern associated with high rates of clinician burnout and with hundreds of billions of dollars in estimated annual administrative waste [1, 2].

The root cause is structural fragmentation. Each of the four principal stakeholders in a patient’s care trajectory, the patient, their physician, the hospital, and the employer or insurer, operates within a separate software ecosystem with no shared intelligence layer. A single clinical note simultaneously generates downstream work across all four. Synthure explores a different shape for this problem: a single grounded pipeline that produces four role specific portals from one note, connected by one shared encounter, so that the same facts and the same actions are reflected everywhere at once.

The contribution of this report is the architecture and an honest, reproducible evaluation of the

parts that are currently testable, together with a plan to evaluate the rest.

2 Architecture

Synthure is organized as a pipeline of specialized agents that feed four interconnected portals:

- **Quality gate**, validates ICD 10 and CPT code formats and deduplicates input
- **Biomedical NER**, extracts entities and maps diagnoses to ICD 10 and procedures to CPT
- **Knowledge resolution**, attaches human readable labels to codes
- **Risk scoring**, computes deterministic denial and readmission estimates from the note text
- **Four writer agents**, produce role specific reports for the patient, physician, hospital, and employer
- **Verification and orchestration**, audit the reports against the extracted facts and connect them

The novel element is the shared encounter. In the shipped demo this is a single mutable state object that every portal reads from and writes to. Approving a prior authorization in the clinician console lowers the denial estimate, advances the claim, and flips the patient view to covered; toggling a code or removing a procedure recomputes the patient cost, the expected reimbursement, and the employer cohort. Every action emits a cross portal event, and the portals carry an inbox so any one can message any other. We verified this propagation with a headless reducer test covering the prior authorization, code edit, financial assistance, claim submission, and messaging flows.

The shipped demo runs entirely in Next.js route handlers with the shared encounter held client side. The production design, where the encounter becomes a server authoritative, event sourced aggregate broadcast over web sockets for multi user, cross device synchronization, is listed as future work in Section 5.

3 The Agent Pipeline

Stage 1, Quality gate. All extracted ICD 10 and CPT codes are validated against compiled patterns. A bare five digit number is accepted as a CPT code only when it is a known code or the surrounding text marks it as one, which prevents stray numbers from being billed.

Stage 2, Biomedical NER. When an Anthropic key is present, a Claude agent reads the note and maps diagnoses to ICD 10 and procedures to CPT using a forced `tool_use` schema, including diagnoses written in plain language or as abbreviations (for example high blood pressure maps to I10). Without a key, a deterministic extractor uses a regular expression for literal codes plus a medication dictionary. Every returned code is format validated, so an invented or malformed code is dropped.

Stage 3, Knowledge resolution. Codes are attached to human readable labels from a curated dictionary. The production design replaces this with semantic retrieval over the full ICD 10 corpus; see Section 5.

Stage 4, Risk scoring. Denial and readmission scores are deterministic heuristics over the note text. They are clearly labelled as estimates in the product and are not a trained model.

Stage 5, Generation, verification, orchestration. Four writer agents (Claude Haiku) produce role specific reports using forced `tool_use` schemas, grounded strictly in the extracted facts. A

verifier and an orchestrator (Claude Sonnet) audit each statement against the facts and connect the four reports.

4 Evaluation (measured)

We report only results we actually ran and can reproduce.

NER, diagnosis to ICD 10. We evaluate the NER stage on the public dataset Inje/SYMTOMS-COT-ICD10-2024, where each item has a gold ICD 10 code. For each item we construct a short note from its symptom list and stated diagnosis, run the extractor, and check whether it recovers the gold code. On $N = 149$ English items:

Extractor	Category recall	Exact recall	Emits a code
Claude NER (Haiku)	60.4% (90/149)	23.5% (35/149)	92.6%
Deterministic regex	0.0% (0/149)	0.0%	0%

The headline finding is the gap: the deterministic extractor recovers no diagnoses from prose, because prose contains no literal codes, while the Claude NER path recovers the correct ICD 10 chapter most of the time. This is the measured justification for the NER stage.

Honest caveats. N is 149; the dataset is symptom and diagnosis to code, which is somewhat idealized relative to full clinical notes; the gold label is a single code per item; category level recall is lenient and exact ICD 10 mapping is hard. Latency: the deterministic extractor runs in about 0.013 ms per note; the 149 Claude NER calls completed in 56 s at a concurrency of 6. Per stage latency profiling on the deployed pipeline is future work.

5 Evaluation Plan (not yet measured)

The following components are described honestly as not yet benchmarked. For each we give the concrete method we would use to make the claim real.

- **NER at scale.** The current N of 149 is small and the dataset is idealized. Plan: evaluate on de identified real clinical notes (for example n2c2 or i2b2 corpora under their data use agreements) and report entity span F1 and exact and category code accuracy with confidence intervals.
- **Denial prediction.** There is no real denial outcome label available, and the demo uses a heuristic. Plan: obtain a claims dataset with adjudication outcomes (paid or denied), train a classifier, and report AUC, precision, recall, and calibration against a held out test set, with a rule based baseline.
- **Retrieval (RAG).** The demo resolves codes from a curated dictionary, not a learned retriever. Plan: embed the full ICD 10 CM corpus into a vector index, assemble a labelled query set (symptom or description to code), and report MRR at k and recall at k against a lexical baseline.
- **Readmission.** The demo uses a heuristic. Plan: calibrate against published CMS HRRP condition rates and report mean absolute error and risk tier agreement.
- **End to end latency.** Plan: instrument the deployed pipeline and report p50, p95, and p99 over real traffic, per stage.

- **Grounding and hallucination.** Plan: sample generated outputs, have reviewers check every statement against the extracted facts, and report a fabrication rate with N and inter rater agreement, alongside the automated code validation already in place.

6 Safety and Grounding

Two properties are enforced by construction rather than asserted by a benchmark.

Grounding. Writer agents receive only the extracted facts and are instructed to use nothing else; the `tool_use` schema forces structured output, and a validation pass drops any code that fails format checks. This makes fabrication of codes and clinical values structurally unlikely, though the planned human review in Section 5 is needed to put a real number on the residual rate.

Three tier autonomy. The system distinguishes actions that may be fully automated (drafting a prior authorization packet, preparing an appeal letter, generating patient education) from actions that require a single human approval, from actions that are never automated (prescribing, diagnosis, treatment changes). The third tier is a design prohibition, not a configuration option.

7 Conclusion

Synthure shows that a single grounded agent pipeline can drive four interconnected stakeholder portals from one clinical note, with actions in any portal rippling through the others in real time, on openly available model APIs and serverless infrastructure. The honest evaluation is narrow but real: Claude based NER recovers the correct ICD 10 category in 60.4% of cases and the exact code in 23.5% (N = 149), against 0.0% for the deterministic baseline on prose. The remaining components are presented as heuristics or design with a concrete plan to evaluate each. We prefer a smaller set of claims we can stand behind to a larger set we cannot reproduce.

References

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